60 Tricky Coding Problems for Freshers (Java/Python/C++)

A curated list of interview-friendly coding problems frequently asked in MNC hiring assessments.

Arrays

1. Find Missing Number
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

2. Find Duplicate Number
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

3. Move All Zeros to End
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

4. Second Largest Element
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

5. Leaders in an Array
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

6. Buildings with Ocean View
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

7. Equilibrium Index
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

8. Product of Array Except Self
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

Strings

9. First Non-Repeating Character
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

10. Check Anagram
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

11. Reverse Words
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

12. Longest Substring Without Repeating Characters
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

13. Valid Parentheses
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

14. String Compression
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

Searching & Sorting

15. Binary Search
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

16. Merge Two Sorted Arrays
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

17. Quick Sort
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

18. Find Kth Largest Element
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

19. Find Majority Element
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

20. Peak Element
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

HashMap / HashSet

21. Two Sum
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

22. Count Frequency
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

23. Duplicate Transaction Detection
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

24. Find Common Elements of Two Arrays
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

25. Longest Consecutive Sequence
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

Stack & Queue

26. Next Greater Element
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

27. Stock Span Problem
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

28. Min Stack
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

29. Reverse Queue
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

30. Circular Queue
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

Linked List

31. Reverse Linked List
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

32. Detect Loop
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

33. Middle Node
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

34. Remove Nth Node From End
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

35. Merge Two Sorted Lists
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

Scenario Based

36. Hospital Emergency Queue
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

37. Flight Departure Board
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

38. Student Race Ranking
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

39. Product Warehouse
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

40. Shopping Mall Sales Analytics
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

41. Traffic Analysis
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

42. Rain Water Trapping
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

43. Buildings Visible From Left
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

44. Buildings Visible From Right
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

45. Count Visible Buildings
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

Medium Level

46. Rotate Array by K Positions
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

47. Spiral Matrix Traversal
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

48. Matrix Rotation (90°)
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

49. Search in Rotated Sorted Array
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

50. Merge Intervals
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

Bonus

51. Celebrity Problem
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

52. Gas Station Circular Tour
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

53. Container With Most Water
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

54. Trapping Rain Water
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

55. Largest Rectangle in Histogram
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

56. Sliding Window Maximum
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

57. Minimum Window Substring
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

58. LRU Cache Design
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

59. Top K Frequent Elements
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.

60. Task Scheduler
Practice goal: Understand the problem, identify the optimal data structure/algorithm, analyze time complexity, and implement a clean solution.